

1. Get and align coefficients

This table shows, for each feature, how it affects Log_Price.

```
# Get feature names
feature_names = X_train.columns

# Coefficients
lin_coef = model.coef_
ridge_coef = ridge_model.coef_
lasso_coef = lasso_model.coef_

coef_df = pd.DataFrame({
    'Feature': feature_names,
    'Linear_Coeff': lin_coef,
    'Ridge_Coeff': ridge_coef,
    'Lasso_Coeff': lasso_coef
})

coef_df
```

	Feature	Linear_Coeff	Ridge_Coeff	Lasso_Coeff	
0	Brand	-0.056075	-0.055999	-0.013741	
1	Price	0.000017	0.000017	0.000019	
2	Body	-0.078491	-0.078477	-0.039125	
3	Mileage	0.000208	0.000208	-0.000121	
4	EngineV	0.004920	0.004915	0.000000	
5	Engine Type	-0.011897	-0.011807	0.000000	
6	Registration	0.766631	0.763506	0.000000	
7	Year	0.067574	0.067593	0.064858	

Next steps:

[Generate code with coef_df](#)[New interactive sheet](#)

2. How to interpret a coefficient (for log price)

Your target is `Log_Price`, so coefficients are in log scale.

For a numeric feature, say `Year`:

- Suppose Linear coeff for `Year` is `0.06`.
- That means: for +1 year (keeping other features constant),

log price increases by `0.06`.

- Convert to percentage effect on price

```
coef_df['Linear_%_Change'] = (np.exp(coef_df['Linear_Coeff']) - 1) * 100
coef_df[['Feature', 'Linear_Coeff', 'Linear_%_Change']]
```

	Feature	Linear_Coeff	Linear_%_Change	
0	Brand	-0.056075	-5.453174	
1	Price	0.000017	0.001668	
2	Body	-0.078491	-7.548956	
3	Mileage	0.000208	0.020789	
4	EngineV	0.004920	0.493238	
5	Engine Type	-0.011897	-1.182617	
6	Registration	0.766631	115.250167	
7	Year	0.067574	6.990894	

Interpretation example:

- `Mileage` coeff = `-0.0004`

→ For +1 km, price drops ≈

$(e^{-0.0004} - 1) \times 100 \approx -0.04\%$

→ For +10,000 km, multiply by 10,000 (roughly -40% price impact, depending on exact value).

For encoded categorical features (Brand, Body, Engine Type, Registration):

- They're LabelEncoded integers, so the numeric order is artificial.
- Coefficients are harder to interpret directly (0→1→2 isn't a natural order).
- Better story: "This feature matters, but to interpret categories, we'd need One-Hot Encoding."

For now you can just say: non-zero coefficients here indicate brand/body/engine type influence price.